Assignment 2 Solutions - Condensed Version

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Problem 1: Probability with Dice

(a) At least one 6 in 4 rolls:

Complement approach:

$$\begin{aligned}P(\text{no 6 in 4 rolls}) &= \left(\frac{5}{6}\right)^4 \\ &= \frac{5^4}{6^4} = \frac{625}{1296}\end{aligned}$$

Therefore:

$$\begin{aligned}P(\text{at least one 6}) &= 1 - \frac{625}{1296} \\ &= \frac{1296-625}{1296} = \frac{671}{1296} \\ &= \boxed{0.5177}\end{aligned}$$

(b) At least two 6s (binomial $n = 4, p = 1/6$):

Use complement: $P(X \geq 2) = 1 - [P(X = 0) + P(X = 1)]$

For $X = 0$:

$$\begin{aligned}P(X=0) &= \binom{4}{0}\left(\frac{1}{6}\right)^0\left(\frac{5}{6}\right)^4 \\ &= 1 \times 1 \times \frac{625}{1296} = \frac{625}{1296}\end{aligned}$$

For $X = 1$:

$$\begin{aligned}P(X=1) &= \binom{4}{1}\left(\frac{1}{6}\right)^1\left(\frac{5}{6}\right)^3 \\ &= 4 \times \frac{1}{6} \times \frac{125}{216} \\ &= \frac{500}{1296}\end{aligned}$$

Sum: $P(X = 0) + P(X = 1) = \frac{625+500}{1296} = \frac{1125}{1296}$

Therefore:

$$\begin{aligned}P(X \geq 2) &= 1 - \frac{1125}{1296} = \frac{171}{1296} \\ &= \boxed{0.1319}\end{aligned}$$

(c) Expected number of 6s:

For binomial distribution:

$$\begin{aligned}E(X) &= np = 4 \times \frac{1}{6} \\ &= \frac{4}{6} = \frac{2}{3} \\ &= \boxed{0.6667}\end{aligned}$$

Problem 2: Conditional Probability

Given: $P(A) = 0.6$, $P(B) = 0.4$, $P(A \cap B) = 0.2$

(a) Union probability:

Using addition rule:

$$\begin{aligned}P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.6 + 0.4 - 0.2 \\ &= \boxed{0.8}\end{aligned}$$

(b) Conditional probability:

By definition:

$$\begin{aligned}P(A|B) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{0.2}{0.4} \\ &= \boxed{0.5}\end{aligned}$$

Interpretation: Given B occurred, probability of A is 0.5.

(c) Test for independence:

Events are independent if $P(A \cap B) = P(A) \cdot P(B)$

Check:

$$\begin{aligned}P(A) \cdot P(B) &= 0.6 \times 0.4 = 0.24 \\ P(A \cap B) &= 0.2\end{aligned}$$

Since $0.24 \neq 0.2$, events are $\boxed{\text{Not independent}}$

Alternative check: $P(A|B) = 0.5 \neq P(A) = 0.6$ confirms dependence.

Problem 3: Bayes' Theorem

Given: $P(D) = 0.01$, $P(+|D) = 0.95$, $P(+|D^c) = 0.05$

Find $P(D|+)$ using Bayes' theorem:

$$P(D|+) = \frac{P(+|D) \cdot P(D)}{P(+)}$$

Step 1: Find $P(+)$ using law of total probability:

$$\begin{aligned}P(+) &= P(+|D)P(D) + P(+|D^c)P(D^c) \\ &= P(+|D)P(D) + P(+|D^c)(1 - P(D)) \\ &= 0.95(0.01) + 0.05(0.99) \\ &= 0.0095 + 0.0495 \\ &= 0.059\end{aligned}$$

Step 2: Apply Bayes' formula:

$$\begin{aligned} P(D|+) &= \frac{P(+|D)\cdot P(D)}{P(+)} \\ &= \frac{0.95\times 0.01}{0.059} \\ &= \frac{0.0095}{0.059} \\ &= \boxed{0.1610} \end{aligned}$$

Interpretation: Only 16.1% of positive tests indicate true disease (due to low prevalence).

Problem 4: Normal Distribution

$X \sim N(100, 15^2)$ with $\mu = 100$, $\sigma = 15$

(a) $P(X > 115)$:

Standardize:

$$Z = \frac{X-\mu}{\sigma} = \frac{115-100}{15} = \frac{15}{15} = 1$$

From standard normal table:

$$\begin{aligned} P(X > 115) &= P(Z > 1) \\ &= 1 - P(Z \le 1) \\ &= 1 - 0.8413 \\ &= \boxed{0.1587} \end{aligned}$$

(b) $P(85 < X < 110)$:

Standardize both bounds:

$$\begin{aligned} Z_1 &= \frac{85-100}{15} = \frac{-15}{15} = -1 \\ Z_2 &= \frac{110-100}{15} = \frac{10}{15} = 0.667 \end{aligned}$$

From table:

$$\begin{aligned} P(85 < X < 110) &= P(-1 < Z < 0.667) \\ &= P(Z < 0.667) - P(Z < -1) \\ &= 0.7475 - 0.1587 \\ &= \boxed{0.5888} \end{aligned}$$

(c) Find x where $P(X < x) = 0.90$:

From table: $P(Z < 1.282) = 0.90$, so $z_{0.90} = 1.282$

Convert back to X scale:

$$\begin{aligned} x &= \mu + z\sigma \\ &= 100 + 1.282(15) \\ &= 100 + 19.23 \\ &= \boxed{119.23} \end{aligned}$$

The 90th percentile is 119.23.

Problem 5: Central Limit Theorem

Given: $n = 50$, $\mu = 50$, $\sigma = 10$

By CLT, sample mean distribution:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) = N\left(50, \frac{100}{50}\right) = N(50, 2)$$

Standard error:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{50}} = \frac{10}{7.071} = 1.414$$

(a) $P(\bar{X} > 52)$:

Standardize:

$$Z = \frac{\bar{X}-\mu}{\sigma_{\bar{X}}} = \frac{52-50}{1.414} = \frac{2}{1.414} = 1.414$$

From table:

$$\begin{aligned} P(\bar{X} > 52) &= P(Z > 1.414) \\ &= 1 - P(Z \le 1.414) \\ &= 1 - 0.9213 \\ &= \boxed{0.0787} \end{aligned}$$

(b) $P(48 < \bar{X} < 51)$:

Standardize:

$$\begin{aligned} Z_1 &= \frac{48-50}{1.414} = \frac{-2}{1.414} = -1.414 \\ Z_2 &= \frac{51-50}{1.414} = \frac{1}{1.414} = 0.707 \end{aligned}$$

From table:

$$\begin{aligned} P(48 < \bar{X} < 51) &= P(-1.414 < Z < 0.707) \\ &= P(Z < 0.707) - P(Z < -1.414) \\ &= 0.7602 - 0.0787 \\ &= \boxed{0.6815} \end{aligned}$$

Problem 6: Hypothesis Testing (One Sample)

Data: $n = 25$, $\bar{x} = 52$, $s = 8$

Test $H_0 : \mu = 50$ vs $H_a : \mu \neq 50$ at $\alpha = 0.05$

Part 1: Test Statistic

Standard error:

$$SE = \frac{s}{\sqrt{n}} = \frac{8}{\sqrt{25}} = \frac{8}{5} = 1.6$$

t-statistic:

$$\begin{aligned} t &= \frac{\bar{x}-\mu_0}{SE} = \frac{\bar{x}-\mu_0}{s/\sqrt{n}} \\ &= \frac{52-50}{1.6} = \frac{2}{1.6} \\ &= \boxed{1.25} \end{aligned}$$

Part 2: Decision

Degrees of freedom: $df = n - 1 = 24$

Critical value (two-tailed): $t_{0.025,24} = 2.064$

Decision rule: Reject H_0 if $|t| > 2.064$

Since $|t| = 1.25 < 2.064$:

$\boxed{\text{Fail to reject } H_0}$

Not enough evidence that $\mu \neq 50$.

Part 3: p-value

For $t = 1.25$ with $df = 24$ (two-tailed):

From t-table: $0.20 < p < 0.30$

More precisely: $\boxed{p \approx 0.223}$

Since $p > 0.05$, fail to reject H_0.

Part 4: Confidence Interval

95% CI formula:

$$\bar{x} \pm t_{0.025,24} \cdot \frac{s}{\sqrt{n}}$$

Calculate:

$$\begin{aligned} CI &= 52 \pm 2.064(1.6) \\ &= 52 \pm 3.302 \\ &= \boxed{(48.698, 55.302)} \end{aligned}$$

Since $\mu_0 = 50$ is in the CI, we fail to reject H_0 (consistent with test).

Problem 7: Two-Sample t-test

Group A: $n_1 = 30$, $\bar{x}_1 = 75$, $s_1 = 10$

Group B: $n_2 = 35$, $\bar{x}_2 = 70$, $s_2 = 12$

Test $H_0 : \mu_1 = \mu_2$ vs $H_a : \mu_1 \neq \mu_2$

Step 1: Pooled Variance

Formula:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Calculate:

$$\begin{aligned} s_p^2 &= \frac{(30-1)(10)^2 + (35-1)(12)^2}{30 + 35 - 2} \\ &= \frac{29(100) + 34(144)}{63} \\ &= \frac{2900 + 4896}{63} = \frac{7796}{63} \\ &= 123.746 \end{aligned}$$

Pooled SD: $s_p = \sqrt{123.746} = 11.124$

Step 2: Standard Error

Formula:

$$SE = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Calculate:

$$\begin{aligned} SE &= 11.124 \sqrt{\frac{1}{30} + \frac{1}{35}} \\ &= 11.124\sqrt{0.0333 + 0.0286} \\ &= 11.124\sqrt{0.0619} \\ &= 11.124 \times 0.2478 = 2.756 \end{aligned}$$

Step 3: Test Statistic

$$\begin{aligned} t &= \frac{\bar{x}_1 - \bar{x}_2}{SE} = \frac{75 - 70}{2.756} \\ &= \frac{5}{2.756} = \boxed{1.814} \end{aligned}$$

Step 4: Decision

$df = n_1 + n_2 - 2 = 63$

Critical value: $t_{0.05,63} \approx 1.669$ (one-tailed at $\alpha = 0.05$)

Since $t = 1.814 > 1.669$:

$\boxed{\text{Reject } H_0}$ (significant at $\alpha = 0.05$)

Group A mean significantly higher than Group B.

Problem 8: Chi-Square Test

Test if 4 categories are equally distributed.

Observed frequencies: $O = [30, 25, 20, 25]$

Total: $n = 30 + 25 + 20 + 25 = 100$

Expected (equal distribution): $E_i = n/4 = 100/4 = 25$ for each

Chi-Square Calculation:

Formula: $\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$

For each category:

$$\begin{aligned} \text{Cat 1:} \quad & \frac{(30-25)^2}{25} = \frac{25}{25} = 1.0 \\ \text{Cat 2:} \quad & \frac{(25-25)^2}{25} = \frac{0}{25} = 0.0 \\ \text{Cat 3:} \quad & \frac{(20-25)^2}{25} = \frac{25}{25} = 1.0 \\ \text{Cat 4:} \quad & \frac{(25-25)^2}{25} = \frac{0}{25} = 0.0 \end{aligned}$$

Sum: $\chi^2 = 1.0 + 0.0 + 1.0 + 0.0 = \boxed{2.0}$

Decision:

Degrees of freedom: $df = k - 1 = 4 - 1 = 3$

Critical value: $\chi^2_{0.05,3} = 7.815$

Since $\chi^2 = 2.0 < 7.815$:

$\boxed{\text{Fail to reject } H_0}$

Data is consistent with equal distribution across categories.

Problem 9: ANOVA

Three groups, test $H_0 : \mu_1 = \mu_2 = \mu_3$

Part 1: Summary Statistics

Group 1: $n_1 = 5$, $\bar{x}_1 = 12$, $s_1^2 = 10$

Group 2: $n_2 = 5$, $\bar{x}_2 = 15$, $s_2^2 = 12$

Group 3: $n_3 = 5$, $\bar{x}_3 = 18$, $s_3^2 = 8$

Total: $N = 15$

Grand mean:

$$\begin{aligned}\bar{x} &= \frac{\sum n_i \bar{x}_i}{N} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + n_3\bar{x}_3}{N} \\ &= \frac{5(12) + 5(15) + 5(18)}{15} \\ &= \frac{60 + 75 + 90}{15} = \frac{225}{15} \\ &= \boxed{15}\end{aligned}$$

Part 2: Sum of Squares

Between-group SS:

$$\begin{aligned}SS_{Between} &= \sum n_i(\bar{x}_i - \bar{x})^2 \\ &= 5(12-15)^2 + 5(15-15)^2 + 5(18-15)^2 \\ &= 5(-3)^2 + 5(0)^2 + 5(3)^2 \\ &= 5(9) + 0 + 5(9) \\ &= 45 + 45 = \boxed{90}\end{aligned}$$

Within-group SS:

$$\begin{aligned}SS_{Within} &= \sum (n_i - 1)s_i^2 \\ &= (5-1)(10) + (5-1)(12) + (5-1)(8) \\ &= 4(10) + 4(12) + 4(8) \\ &= 40 + 48 + 32 = \boxed{120}\end{aligned}$$

Total SS:

$$SS_{Total} = SS_{Between} + SS_{Within} = 90 + 120 = \boxed{210}$$

Part 3: ANOVA Table

Degrees of freedom:

$$\begin{aligned}df_{Between} &= k - 1 = 3 - 1 = 2 \\ df_{Within} &= N - k = 15 - 3 = 12 \\ df_{Total} &= N - 1 = 14\end{aligned}$$

Mean squares:

$$\begin{aligned}MS_{Between} &= \frac{SS_{Between}}{df_{Between}} = \frac{90}{2} = 45 \\ MS_{Within} &= \frac{SS_{Within}}{df_{Within}} = \frac{120}{12} = 10\end{aligned}$$

Source	SS	df	MS	F
Between	90	2	45	4.5
Within	120	12	10	–
Total	210	14	–	–

F-statistic:

$$F = \frac{MS_{Between}}{MS_{Within}} = \frac{45}{10} = \boxed{4.5}$$

Critical value: $F_{0.05,2,12} = 3.89$

Since $F = 4.5 > 3.89$: $\boxed{\text{Reject } H_0}$

At least one group mean differs significantly.

Problem 10: Simple Linear Regression

Data: $n = 5$, (x, y): $(0, 1.0), (1, 2.9), (2, 5.1), (3, 5.9), (4, 9.2)$

Model: $y = \beta_0 + \beta_1 x + \varepsilon$

Part 1: OLS Estimates

Summary statistics:

$$\begin{aligned}\bar{x} &= \frac{0+1+2+3+4}{5} = \frac{10}{5} = 2 \\ \bar{y} &= \frac{1.0 + 2.9 + 5.1 + 5.9 + 9.2}{5} = \frac{24.1}{5} = 4.82\end{aligned}$$

Sum of squares for X:

$$\begin{aligned}S_{xx} &= \sum (x_i - \bar{x})^2 \\ &= (-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2 \\ &= 4 + 1 + 0 + 1 + 4 = 10\end{aligned}$$

Sum of cross-products:

$$\begin{aligned}S_{xy} &= \sum (x_i - \bar{x})(y_i - \bar{y}) \\ &= (-2)(-3.82) + (-1)(-1.92) \\ &\quad + 0(0.28) + 1(1.08) + 2(4.38) \\ &= 7.64 + 1.92 + 0 + 1.08 + 8.76 \\ &= 19.4\end{aligned}$$

Slope estimate:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{19.4}{10} = \boxed{1.94}$$

Intercept estimate:

$$\begin{aligned}\hat{\beta}_0 &= \bar{y} - \hat{\beta}_1\bar{x}\\ &= 4.82 - 1.94(2)\\ &= 4.82 - 3.88 = \boxed{0.94}\end{aligned}$$

Fitted regression line: $\hat{y} = 0.94 + 1.94x$

Part 2: Fit Statistics

Calculate fitted values and residuals:

x	y	$\hat{y}$	e	e^2
0	1.0	0.94	0.06	0.0036
1	2.9	2.88	0.02	0.0004
2	5.1	4.82	0.28	0.0784
3	5.9	6.76	-0.86	0.7396
4	9.2	8.70	0.50	0.2500
			Sum:	1.072

Error sum of squares:

$$SSE = \sum e_i^2 = \boxed{1.072}$$

Total sum of squares:

$$S_{yy} = \sum(y_i - \bar{y})^2 = 38.708$$

Regression sum of squares:

$$\begin{aligned}SSR &= S_{yy} - SSE\\ &= 38.708 - 1.072 = 37.636\end{aligned}$$

Coefficient of determination:

$$\begin{aligned}R^2 &= \frac{SSR}{S_{yy}} = \frac{37.636}{38.708}\\ &= \boxed{0.9723}\end{aligned}$$

Residual standard error:

$$\begin{aligned}s &= \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{1.072}{3}}\\ &= \sqrt{0.357} = \boxed{0.598}\end{aligned}$$

97.23% of variance in y is explained by x.

Part 3: Hypothesis Test for β_1

Test $H_0 : \beta_1 = 0$ vs $H_a : \beta_1 \neq 0$ at $\alpha = 0.05$

Standard error of slope:

$$\begin{aligned}SE(\hat{\beta}_1) &= \frac{s}{\sqrt{S_{xx}}} = \frac{0.598}{\sqrt{10}}\\ &= \frac{0.598}{3.162} = \boxed{0.189}\end{aligned}$$

t-statistic:

$$\begin{aligned}t &= \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} = \frac{1.94}{0.189}\\ &= \boxed{10.265}\end{aligned}$$

Degrees of freedom: $df = n - 2 = 3$

Critical value: $t_{0.025,3} = 3.182$ (two-tailed)

Since $|t| = 10.265 > 3.182$:

$\boxed{\text{Reject } H_0}$

p-value ≈ 0.002 (highly significant)

95% confidence interval for β_1:

$$\begin{aligned}CI &= \hat{\beta}_1 \pm t_{0.025,3} \cdot SE(\hat{\beta}_1)\\ &= 1.94 \pm 3.182(0.189)\\ &= 1.94 \pm 0.601\\ &= \boxed{(1.339, 2.541)}\end{aligned}$$

Strong evidence of positive linear relationship.

Part 4: Prediction Interval at $x_0 = 5$

Point prediction:

$$\begin{aligned}\hat{y}_0 &= \hat{\beta}_0 + \hat{\beta}_1 x_0\\ &= 0.94 + 1.94(5)\\ &= 0.94 + 9.70 = \boxed{10.64}\end{aligned}$$

Standard error for prediction:

$$\begin{aligned}SE_{pred} &= s\sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}\\ &= 0.598\sqrt{1 + \frac{1}{5} + \frac{(5-2)^2}{10}}\\ &= 0.598\sqrt{1 + 0.2 + \frac{9}{10}}\\ &= 0.598\sqrt{1 + 0.2 + 0.9}\\ &= 0.598\sqrt{2.1} = 0.598(1.449)\\ &= \boxed{0.866}\end{aligned}$$

Variance components:

- Individual variation: 1.0 (47.6%)
- Sampling: $1/5 = 0.2$ (9.5%)
- Distance: $9/10 = 0.9$ (42.9%)

95% prediction interval:

$$\begin{aligned}PI &= \hat{y}_0 \pm t_{0.025,3} \cdot SE_{pred}\\ &= 10.64 \pm 3.182(0.866)\\ &= 10.64 \pm 2.76\\ &= \boxed{(7.88, 13.40)}\end{aligned}$$

Important Note: $x_0 = 5$ is extrapolation (outside observed range $[0, 4]$). Prediction assumes linearity continues beyond data.